

Performance Testing

Date	05 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Browser-based Functional Testing (Flask Local Server)
Type of Testing	Functional Performance Check
Target Module / API	Crop Recommendation Module
Test Environment	Local System (Python Flask Application)
Test Date	05-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Valid crop prediction request	1	30	Correct crop recommendation displayed
2	Multiple prediction attempts	1	60	Application responds consistently
3	Continuous prediction testing	1	120	Stable application performance

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.5 seconds	Pass	Quick response observed
2	Response Time (Max)	< 5 seconds	3.2 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	>1 Req/sec	2 Req/sec	Pass	Suitable for local testing
4	Error Rate	< 1%	0%	Pass	No runtime errors observed
5	CPU Utilization	< 80%	42%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	55%	Pass	Stable memory usage

Step 4: Observations & Analysis

- The application successfully generated crop recommendations for valid inputs.
- Response time remained within acceptable limits during local execution.
- No crashes or unexpected errors were observed.
- Resource utilization remained stable during testing.
- The application is suitable for small-scale deployment and future enhancements.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

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Nitrogen (N)

Phosphorus (P)

Potassium (K)

Temperature (°C)

Humidity (%)

pH

Rainfall (mm)

Predict Crop

OptiCrop

Smart Agricultural Production Optimization Engine

Nitrogen (N)

90

Phosphorus (P)

34

Potassium (K)

57

Temperature (°C)

36

Humidity (%)

83

pH

6.5

Rainfall (mm)

210

Predict Crop

